Homework 31: System of Differential Equations Using Diagonalization

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BCS A, E, F, G

Book: David C. Lay's Linear Algebra Fourth Edition

Section 5.7: Application to Differential Equations (Page 315)

Topic: Decoupling a Dynamical System – *Change of Variable*

Homework 25 – Part b. Find the solution of the following system using eigenvalues and eigenvectors

$$\frac{dx_1}{dt} = -4x_1 + 2x_2, \qquad \frac{dx_2}{dt} = 3x_1 + x_2\,,$$

subjected to initial conditions $x_1(0) = -1$ and $x_2(0) = 4$. Also find the steady state solution of the system.

Solution. Recall from homework's solution that the eigenvalues and corresponding eigenvectors to $A = \begin{bmatrix} -4 & 2 \\ 3 & 1 \end{bmatrix}$ are

$$\lambda_1 = 2, \qquad \lambda_2 = 5, \qquad v_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \qquad v_2 = \begin{bmatrix} -2 \\ 1 \end{bmatrix}.$$

Henceforth

$$D = \begin{bmatrix} 2 & 0 \\ 0 & -5 \end{bmatrix}, \qquad P = \begin{bmatrix} 1 & -2 \\ 3 & 1 \end{bmatrix}.$$

Using change of variables, one can write $y' = Dy$. That is

$$\begin{bmatrix} y_1' \\ y_2' \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & -5 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 2y_1 \\ -5y_2 \end{bmatrix}.$$

So the equations are

$$y_1' = 2y_1, \qquad y_2' = -5y_2\,.$$

Integrating both of the equations give

$$y_1 = c_1 e^{2t}, \qquad y_2 = c_2 e^{-5t}\,.$$

One has

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} c_1 e^{2t} \\ c_2 e^{-5t} \end{bmatrix}.$$

Since

$$x = Py\,.$$

This implies

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} c_1 e^{2t} \\ c_2 e^{-5t} \end{bmatrix} = \begin{bmatrix} c_1 e^{2t} - 2c_2 e^{-5t} \\ 3c_1 e^{2t} + c_2 e^{-5t} \end{bmatrix} = \begin{bmatrix} c_1 e^{2t} \\ 3c_1 e^{2t} \end{bmatrix} + \begin{bmatrix} -2c_2 e^{-5t} \\ c_2 e^{-5t} \end{bmatrix} = c_1 e^{2t} \begin{bmatrix} 1 \\ 3 \end{bmatrix} + c_2 e^{-5t} \begin{bmatrix} -2 \\ 1 \end{bmatrix}.$$

Hence

$$x = c_1 e^{\lambda_1 t} v_1 + c_2 e^{\lambda_2 t} v_2\,.$$

Problem. Find the solution of the following system using eigenvalues and eigenvectors

$$\frac{dx_1}{dt} = x_1 + 2x_2, \qquad \frac{dx_2}{dt} = 2x_1 + 4x_2\,,$$

subjected to initial conditions $x_1(0) = -1$ and $x_2(0) = 4$. Also find the steady state solution of the system.

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Good Luck